

Sri Lanka Institute of Information Technology



Faculty of Computing

IT2011 - Artificial Intelligence and Machine Learning

Lab 05 - Introduction to Generative AI

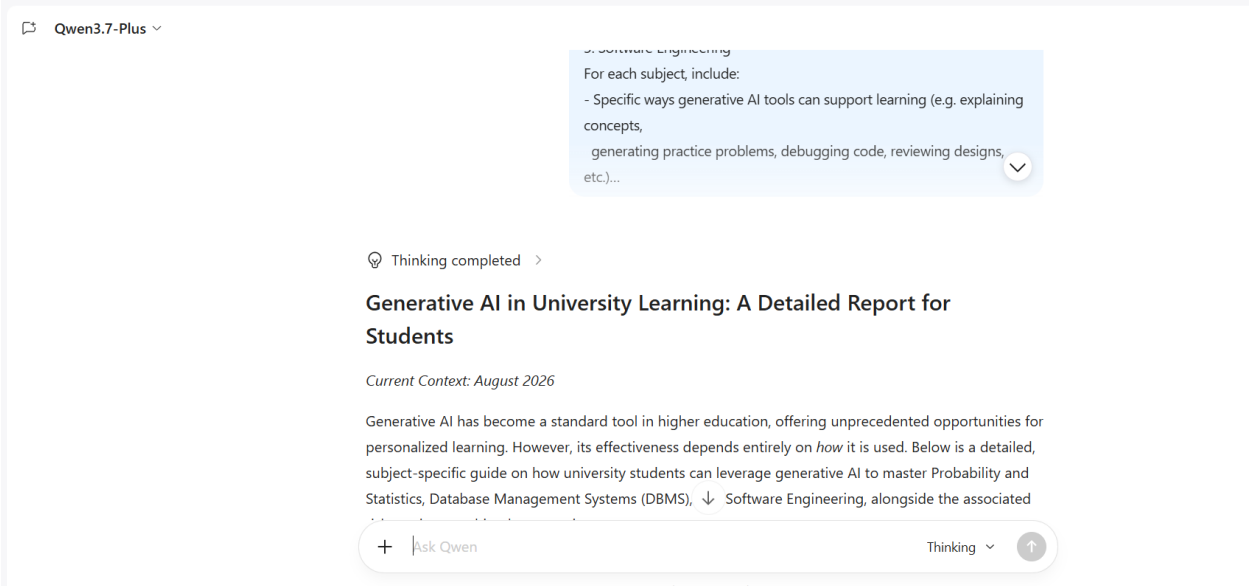
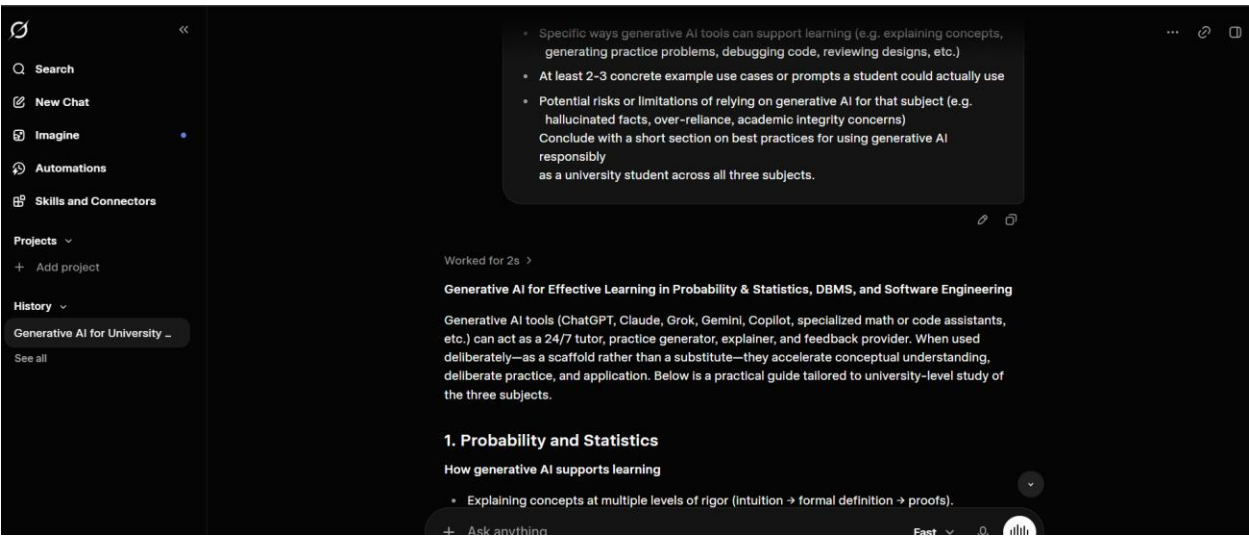
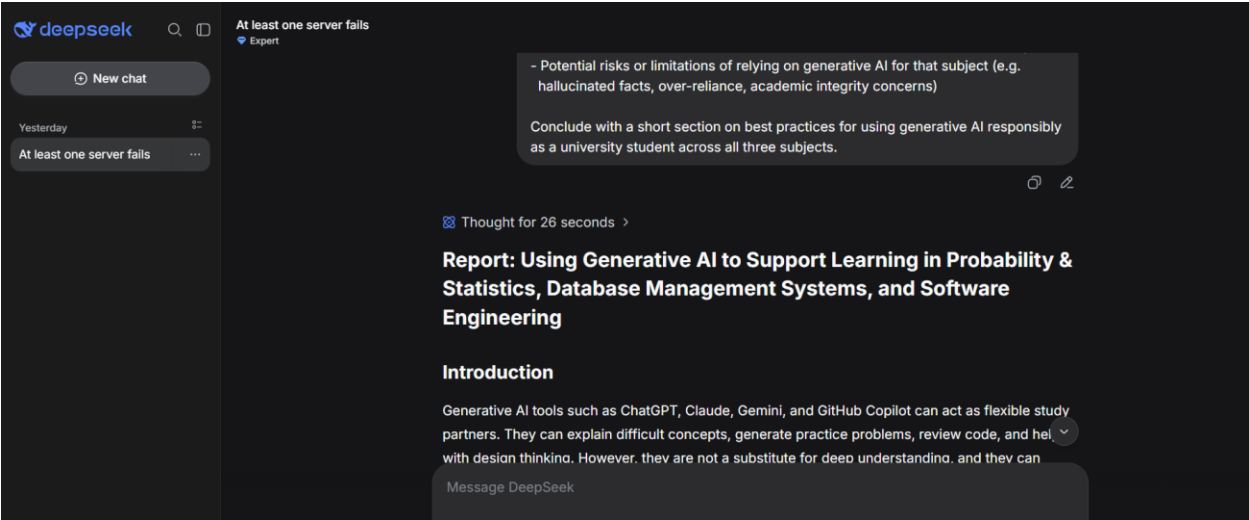
Submission: Exercises 2 to 6

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Exercise 3: Generative AI for Learning

LLMs used: DeepSeek R1, Qwen (Thinking mode), Grok (DeepSearch).

Prompt: "Generate a detailed report on how a university student can use generative AI to learn Probability & Statistics, DBMS, and Software Engineering - including specific uses, example prompts, risks, and best practices."



All three models produced full reports covering: explaining concepts, generating practice problems/SQL/code, debugging, and reviewing designs - for each subject - plus risks (hallucinations, over-reliance, academic integrity) and a best-practices summary (verify outputs, attempt problems first, disclose AI use per university policy).

Reflection: Grok's DeepSearch report was the most detailed, going deepest into concrete technical examples (isolation levels, SOLID principles). Qwen's standout idea was framing AI as a "Socratic tutor" rather than an answer key. DeepSeek was the most concise. Overall, Grok's report was the most useful, but Qwen's prompting approach is the most practically applicable habit going forward.